

**DESIGN AND DEVELOPMENT OF A GLOBAL MULTIDIMENSIONAL TALENT
MARKETPLACE AND CASTING MANAGEMENT SYSTEM FOR MODELS,
INDUSTRY PROFESSIONALS, AND PAGEANT ORGANIZERS**

By

A.M.S.P Athapaththu

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ABSTRACT

The rapid evolution of Information and Communication Technology (ICT) has significantly transformed recruitment and talent management across many industries. While digital recruitment platforms have become increasingly common in corporate environments, the global fashion, entertainment, and pageantry industries continue to rely heavily on fragmented recruitment methods such as social media platforms, independent agencies, and informal professional networks. These approaches often lack profile verification, secure communication, centralized portfolio management, and efficient talent discovery mechanisms.

This research proposes the design and development of a Global Multidimensional Talent Marketplace and Casting Management System that integrates models, industry professionals, and pageant organizers within a single digital ecosystem. The proposed platform provides secure multi-role authentication, professional portfolio management, country-based talent discovery, centralized casting advertisements, recruiter dashboards, and pageant management modules. The system aims to improve transparency, efficiency, and trust while reducing the administrative challenges associated with international talent recruitment.

The study adopts the Design Science Research (DSR) methodology to analyze stakeholder requirements, design the software architecture, develop a prototype, and evaluate system usability and performance. Primary data will be collected through interviews and questionnaires involving models, recruiters, and pageant organizers, while secondary data will be gathered from scholarly journals and industry reports. The expected outcome is a scalable software solution that improves recruitment workflows and supports global collaboration within the creative industry.

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Table of Contents

CHAPTER 01 – INTRODUCTION

- 1.1 Introduction**
- 1.2 Background of the Study**
- 1.3 Problem Statement**
- 1.4 Research Questions**
- 1.5 Aim of the Research**
- 1.6 Research Objectives**
- 1.7 Scope of the Study**
- 1.8 Significance of the Study**
- 1.9 Target Users and Stakeholders**
- 1.10 Overview of the Proposed System**
- 1.11 Chapter Summary**

CHAPTER 02 – LITERATURE REVIEW

- 2.1 Introduction**
- 2.2 Overview of Digital Talent Recruitment**
 - 2.3 Existing Talent Recruitment Platforms**
 - 2.3.1 General Professional Networking Platforms**
 - 2.3.2 Freelancing Platforms**
 - 2.3.3 Social Media-Based Recruitment**
 - 2.3.4 Modeling and Casting Platforms**
- 2.4 Digital Portfolio Management**
- 2.5 Online Casting Management Systems**
- 2.6 Talent Search and Filtering**
- 2.7 Recommendation and Matching Systems**
- 2.8 Profile Verification and Trust**
- 2.9 Role-Based Access Control and Security**
- 2.10 Communication and Collaboration Features**
- 2.11 Comparison of Existing Systems**
- 2.12 Research Gap**
- 2.13 Theoretical Foundation**
- 2.14 Conceptual Framework**
- 2.15 Chapter Summary**

CHAPTER 03 – METHODOLOGY

3.1 Introduction

3.2 Research Design

3.3 Research Approach

3.4 Data Collection Methods

3.5 Requirement Gathering and Analysis

3.6 System Development Methodology

3.6.1 Agile Development Methodology

3.6.2 Sprint Planning

3.6.3 Iterative Development

3.6.4 Testing and Feedback

3.7 System Requirements

3.7.1 Functional Requirements

3.7.2 Non-Functional Requirements

3.8 System Design

3.8.1 System Architecture

3.8.2 Use Case Diagram

3.8.3 Class Diagram

3.8.4 Entity Relationship Diagram

3.8.5 Sequence Diagrams

3.8.6 Activity Diagrams

3.8.7 Database Design

3.9 System Implementation

3.9.1 Frontend Technologies

3.9.2 Backend Technologies

3.9.3 Database Technologies

3.9.4 Development Tools and Frameworks

3.9.5 Cloud Deployment

3.10 System Testing

3.10.1 Unit Testing

3.10.2 Integration Testing

3.10.3 System Testing

3.10.4 User Acceptance Testing

3.10.5 Performance Testing

3.10.6 Security Testing

3.11 System Security

3.12 Deployment

3.13 Ethical Considerations

3.14 Research Constraints and Limitations

3.15 Chapter Summary

CHAPTER 04 – RESULTS

- 4.1 Introduction**
- 4.2 System Implementation Results**
 - 4.2.1 Authentication and User Management**
 - 4.2.2 Talent Profile and Portfolio Management**
 - 4.2.3 Casting Management**
 - 4.2.4 Search and Filtering**
 - 4.2.5 Messaging**
 - 4.2.6 Administration**
- 4.3 Functional Testing Results**
- 4.4 Integration Testing Results**
- 4.5 Security Testing Results**
- 4.6 Performance Testing Results**
- 4.7 User Acceptance Testing Results**
- 4.8 Deployment Results**
- 4.9 Achievement of Research Objectives**
- 4.10 Chapter Summary**

CHAPTER 05 – DISCUSSION

- 5.1 Introduction**
- 5.2 Discussion of the Main Findings**
- 5.3 Discussion of Authentication and Role-Based Access**
- 5.4 Discussion of Talent Profiles and Portfolios**
- 5.5 Discussion of Casting Management**
- 5.6 Discussion of Search and Talent Discovery**
- 5.7 Discussion of Communication**
- 5.8 Discussion of Security**
- 5.9 Discussion of System Performance and Scalability**
- 5.10 Relationship Between Findings and Research Questions**
- 5.11 Relationship with the Research Gap**
- 5.12 Contribution of the Developed System**
- 5.13 Limitations of the Study**
- 5.14 Recommendations for Future Development**
- 5.15 Chapter Summary**

REFERENCES

CHAPTER 1 – INTRODUCTION

1.1 Chapter Overview

This chapter introduces the research by presenting the background of the study, identifying the research problem, defining the research aim and objectives, formulating the research questions, describing the scope and significance of the study, and outlining the overall thesis structure. It establishes the rationale for developing a centralized digital platform to support global talent recruitment within the fashion, entertainment, and pageantry industries.

1.2 Background of the Study

Information and Communication Technology (ICT) has changed the way organizations communicate, conduct business, and recruit employees. Today, digital platforms help employers connect with talented people from different parts of the world. Technologies such as cloud computing, mobile applications, artificial intelligence, and secure authentication have made recruitment faster, easier, and more efficient.

The fashion and entertainment industries have grown rapidly over the past few years. Fashion shows, advertisements, films, digital content, modeling competitions, and beauty pageants create many job opportunities worldwide. As these industries continue to grow, there is a greater need for better recruitment systems.

In the past, modeling agencies helped connect models with clients. They managed portfolios, negotiated contracts, and guided models in their careers. Today, many models work independently and use social media platforms like Instagram, TikTok, Facebook, and LinkedIn to promote themselves. Although these platforms provide good exposure, they are not designed for professional recruitment. Recruiters often find it difficult to verify profiles, check experience, and identify suitable candidates.

The lack of a centralized recruitment platform creates many challenges. Models struggle to find genuine job opportunities. Recruiters must search through different websites and social media accounts to find suitable talent. Fake profiles, fraudulent casting calls, and poor communication also reduce trust in the recruitment process.

Industry professionals such as fashion brands, modeling agencies, photographers, film directors, commercial directors, and event organizers also face difficulties when searching for qualified talent. Most recruitment is still done through manual portfolio reviews, agency referrals, or social media. These methods are often slow and inefficient.

Pageant organizers face similar problems. They need an organized system to manage applications, communicate with contestants, and conduct recruitment. However, many organizers still rely on emails, social media posts, and manual registration, which increases their workload.

Existing platforms solve only some of these problems. LinkedIn mainly supports corporate recruitment, while Backstage and StarNow focus on specific areas of the entertainment industry. Most of these platforms do not provide one secure system that connects models, recruiters, and pageant organizers. They also lack advanced search, profile verification, and country-based talent discovery features.

Algorithm-based talent matching can further improve the recruitment process by automatically comparing the requirements of a casting opportunity with the characteristics of available talents. The proposed system can consider factors such as age, height, location, experience, category, skills, and portfolio information when identifying suitable candidates. A weighted matching approach can be used to calculate a suitability score for each talent and rank candidates according to their compatibility with a particular casting requirement. This can reduce the time required for manual candidate searching and help recruiters identify relevant talents more efficiently.

The proposed system can also use search and filtering algorithms to support recruiters in finding suitable talents based on specific requirements. Instead of manually reviewing a large number of profiles, recruiters can enter relevant criteria and receive a filtered list of candidates. These algorithm-based approaches can improve the efficiency, consistency, and usability of the recruitment process.

Therefore, there is a need for a specialized digital platform that combines professional talent profiles, portfolio management, casting management, advanced search and filtering, and algorithm-based talent matching within a single system. The proposed Global Multidimensional Talent Marketplace and Casting Management System aims to address these limitations by providing a centralized and structured environment for models, industry professionals, and pageant organizers.

1.3 Problem Statement

1.3.1 General Problem

Despite advances in digital recruitment, the fashion, entertainment, and pageant industries still rely heavily on traditional and fragmented recruitment methods. Many models and talents depend on social media, personal networks, and modeling agencies to find opportunities, while recruiters manually search through different platforms to identify suitable candidates. Existing platforms do

not provide one secure and centralized system that allows models, recruiters, industry professionals, and pageant organizers to work together.

Furthermore, existing recruitment methods provide limited support for verified professional profiles, structured casting management, country-based talent discovery, and automated talent matching. Recruiters often need to manually review a large number of profiles and portfolios, which increases the time and effort required to identify suitable candidates. The absence of algorithm-based matching and recommendation mechanisms also makes it difficult to efficiently compare talent profiles with specific casting requirements.

As a result, the current recruitment process can be time-consuming, inefficient, costly, and less transparent. Therefore, there is a need for a centralized digital platform that integrates talent management, portfolio management, casting management, advanced search and filtering, and algorithm-based talent matching within a single system.

1.3.2 Specific Problems

The research identifies the following specific problems based on the existing literature and analysis of current recruitment practices:

- * Lack of a centralized platform that integrates models, industry professionals, and pageant organizers.
- * Heavy dependence on unverified social media platforms for recruitment.
- * Difficulty verifying professional experience and identity.
- * Limited access to international recruitment opportunities for freelance models.
- * Time-consuming manual portfolio reviews and casting management.
- * Absence of dedicated pageant recruitment modules.
- * Inadequate role-based access control and secure authentication.
- * Limited country-specific search and filtering capabilities.
- * Challenges associated with managing large multimedia portfolios.
- * Insufficient communication tools for recruiters and talent.

- * Lack of an effective algorithm-based mechanism for matching talents with suitable casting opportunities.
- * Limited automated recommendation capabilities for helping recruiters identify suitable candidates.

1.4 Research Questions

Main Research Question

How can a centralized global talent marketplace and casting management system improve recruitment efficiency, transparency, and trust within the fashion, entertainment, and pageantry industries through secure digital recruitment, advanced search, and algorithm-based talent matching?

Specific Research Questions

1. What technical and professional challenges do models, recruiters, and pageant organizers face when using existing recruitment methods?
2. How can a secure multi-role software architecture improve interactions among different stakeholder groups?
3. How can country-based talent discovery and advanced filtering improve international recruitment?
4. How can profile verification reduce fraudulent recruitment activities?
5. How can an algorithm-based matching mechanism identify and rank suitable talents according to casting requirements?
6. How effective is the proposed system in terms of usability, performance, matching accuracy, and user satisfaction?

1.5 Research Motivation

The motivation for this research arises from the increasing demand for reliable and efficient digital recruitment solutions within the creative industry. Although digital technologies have transformed many business sectors, professional recruitment for models and pageant participants remains fragmented and dependent on informal networking, social media, agencies, and manual portfolio reviews.

Recruiters may need to review a large number of talent profiles before identifying suitable candidates for a specific casting opportunity. This creates a need for a more efficient method of searching, comparing, and recommending talents. An algorithm-based matching mechanism can assist recruiters by comparing casting requirements with relevant talent attributes and producing ranked results.

A secure and centralized platform can therefore simplify recruitment, improve transparency, create greater international exposure for freelance professionals, and help recruiters identify qualified candidates more efficiently. The integration of search, filtering, and algorithm-based matching can further reduce manual effort and improve the overall recruitment process.

1.6 Research Aim

The aim of this research is to design and develop a global talent marketplace and casting management system for models, aspiring talents, modeling agencies, photographers, pageant organizers, and other industry professionals.

The proposed system will provide a single web-based platform where users can create professional profiles, manage portfolios, search for talent, and apply for or post casting opportunities. It will help reduce the problems associated with manual recruitment and scattered information while making communication and talent discovery easier.

The system also aims to improve recruitment by providing secure user profiles, advanced search and filtering features, organized casting management, and algorithm-based talent matching and recommendation. The matching mechanism will compare relevant talent attributes with casting requirements to identify and rank potentially suitable candidates.

Overall, the research seeks to create a more efficient, transparent, secure, and accessible digital recruitment platform for the global modeling, entertainment, and pageant industries.

1.7 Research Objectives

1.7.1 General Objective

To design and develop a scalable, secure, and user-friendly global talent marketplace and casting management system with advanced search, filtering, and algorithm-based talent matching capabilities.

1.7.2 Specific Objectives

1. To analyze existing systems and industry practices

Investigate current casting platforms, agency workflows, and talent management tools to identify inefficiencies, limitations, and gaps in functionality.

2. To identify system requirements

Gather functional and non-functional requirements from key stakeholders such as models, agencies, photographers, recruiters, and pageant organizers.

3. To design the system architecture

Develop a structured architecture that supports multi-role users, including modules for user management, casting management, talent portfolios, communication, search, and talent matching.

4. To develop a web-based platform

Implement a functional system that allows users to register, create professional profiles, upload portfolios, and interact with casting opportunities.

5. To implement advanced search and filtering mechanisms

Enable industry professionals to search and filter talents based on attributes such as age, height, experience, location, category, skills, and portfolio information.

6. To design and implement an algorithm-based talent matching mechanism

Develop a matching algorithm that compares talent attributes with casting requirements and calculates a suitability score to identify and rank suitable candidates.

7. To incorporate casting management functionalities

Allow agencies and organizers to post casting calls, manage applications, shortlist candidates, and track selection processes.

8. To implement talent recommendation capabilities

Provide recommendations of potentially suitable talents to recruiters based on relevant profile attributes and casting requirements.

9. To ensure usability, accessibility, and scalability

Design a system that is responsive, easy to use, secure, and capable of supporting global users with minimal performance degradation.

10. To evaluate system effectiveness

Assess the system using functional testing, user testing, feedback collection, performance evaluation, and evaluation of the implemented matching and recommendation mechanisms.

1.8 Scope of the Study

The scope of this study defines the boundaries within which the system is designed and implemented. The study focuses on developing a web-based global talent marketplace and casting management system that supports talent discovery, portfolio management, casting operations, and algorithm-based talent matching.

1.8.1 In-Scope Areas

The following areas are included within the scope of this research:

Development of a web-based global talent marketplace and casting management system.

Inclusion of multiple user roles:

- Models and aspiring talents
- Modeling agencies
- Photographers
- Pageant organizers
- Casting directors and industry professionals

User functionalities such as:

- Registration and authentication
- Profile creation and management
- Portfolio, image, and video uploads
- Casting call creation and application submission
- Talent search and filtering
- Country-based talent discovery
- Messaging and communication between relevant users

- Development of an algorithm-based talent matching mechanism using relevant profile and casting attributes.
- Calculation of a suitability score to compare talent profiles with casting requirements using predefined matching criteria.
- Basic talent recommendation and ranking based on matching results.
- Casting application management, including candidate shortlisting and selection tracking.
- Admin panel for system moderation and management.

1.8.2 Out-of-Scope Areas

The following features are not included in this research:

- Full commercial deployment of the system.
- Advanced artificial intelligence features such as facial recognition, deep learning-based talent analysis, or computer vision-based assessment.
- Machine-learning-based automatic talent scoring.
- Online payment, financial transactions, or digital contract signing.
- Legal identity verification or professional background checks.
- Fully automated decision-making for final talent selection. Final recruitment decisions remain with the relevant recruiter or organizer.
- Mobile application development, which may be considered as future work.

1.8.3 Geographic and Operational Scope

The system is designed for global usage and is not restricted to a specific country or region. It is intended to allow talents and industry professionals from different countries to create profiles, discover opportunities, and participate in casting activities.

However, the implementation and evaluation of the system may be limited to a prototype or academic environment due to the available development time, resources, and access to real-world industry users and datasets.

1.9 Significance of the Study

This research is significant in several key dimensions:

1.9.1 Industry Impact

The modeling, entertainment, and pageant industries often depend on social media, email, agencies, and physical auditions to recruit talent. These methods can be slow, fragmented, and difficult to manage. The proposed system provides a centralized digital platform that brings talents, recruiters, industry professionals, and pageant organizers together.

The inclusion of algorithm-based talent matching can further support recruiters by comparing casting requirements with relevant talent attributes and identifying potentially suitable candidates.

1.9.2 Efficiency Improvement

The proposed system aims to reduce:

- Time spent searching for suitable talent.
- Manual comparison of multiple talent profiles.
- Manual communication between agencies, recruiters, and models.
- Dependency on physical auditions and location-based recruitment.
- Time required to identify suitable candidates for casting opportunities.

The search, filtering, and matching mechanisms can help recruiters narrow down suitable candidates more efficiently.

1.9.3 Talent Visibility and Opportunities

Aspiring models and talents often find it difficult to gain visibility and access international opportunities. The proposed platform can help by:

- Providing users with global exposure.
- Providing direct access to casting opportunities.
- Allowing users to create organized and professional portfolios.
- Helping talents discover opportunities that match their profile attributes.
- Reducing geographical barriers to talent discovery.

1.9.4 Technological Contribution

This study contributes to the development of:

- Digital marketplace systems.
- Talent management platforms.
- Web-based systems with multiple user roles.
- Online casting and recruitment systems supported by databases.
- Algorithm-based talent matching and recommendation mechanisms.
- Search and filtering systems for specialized recruitment.

1.9.5 Academic Contribution

This research can serve as a reference for future studies related to:

- Online recruitment systems.
- Industry-specific marketplace platforms.
- Digital casting and talent management systems.
- Algorithm-based talent matching systems.
- Scalable web application design.
- Multi-role recruitment platforms.

1.10 Thesis Structure

This thesis is organized into five main chapters:

Chapter 1: Introduction

This chapter introduces the research background, identifies the research problem, and presents the research questions, aim, objectives, scope, and significance of the study. It provides an overview of the need for a centralized global talent marketplace and casting management system for the modeling, entertainment, and pageant industries.

Chapter 2: Literature Review

This chapter reviews previous studies, existing systems, technologies, and theoretical concepts related to talent marketplaces, casting platforms, digital recruitment, portfolio management, recommendation systems, and talent matching. It also identifies the limitations of existing systems and establishes the research gap that supports the development of the proposed solution.

Chapter 3: Methodology, System Analysis and Design

This chapter presents the research methodology and software development approach used for the study. It describes requirement analysis, system architecture, database design, algorithms, UML diagrams, system modules, implementation technologies, testing methods, deployment, and security considerations. The chapter also explains the proposed talent matching and recommendation algorithms.

Chapter 4: System Implementation and Results

This chapter describes the implementation of the proposed system and presents the results obtained from system development and evaluation. It includes the implemented system features, user interfaces, algorithm implementation, testing results, performance evaluation, and user feedback.

Chapter 5: Discussion, Conclusion and Future Work

This chapter discusses the findings of the research in relation to the research questions, objectives, and literature review. It evaluates the effectiveness and limitations of the proposed system and provides the conclusion of the study. Recommendations for future improvements, such as advanced AI integration, mobile applications, and machine-learning-based matchmaking, are also presented.

1.11 Chapter Summary

This chapter introduced the proposed Global Multidimensional Talent Marketplace and Casting Management System and established the foundation of the research. It discussed the background of the study, problem statement, research questions, motivation, aim, objectives, scope, and significance.

The chapter identified major challenges in existing talent recruitment practices, including fragmented recruitment platforms, manual portfolio reviews, limited international talent discovery, insufficient communication, and the lack of effective algorithm-based talent matching.

The proposed system aims to address these problems through a centralized web-based platform that provides professional profiles, portfolio management, casting management, advanced search and filtering, and algorithm-based talent matching and recommendation.

The following chapter reviews existing literature, systems, technologies, and research related to digital talent marketplaces, casting management, recommendation systems, and talent matching. It also identifies the research gap that supports the development of the proposed system.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to talent marketplace systems, digital recruitment platforms, casting management systems, portfolio-based applications, recommendation systems, talent matching algorithms, and web-based architectures. The review aims to examine existing approaches, identify their strengths and limitations, and establish the research gap that motivates the development of the proposed Global Talent Marketplace and Casting Management System.

The chapter also discusses modern technologies and concepts such as Role-Based Access Control (RBAC), intelligent recommendation systems, algorithm-based matching, advanced search and filtering, cloud computing, and portfolio management. These concepts are relevant to the proposed system because the platform must support different types of industry users while allowing recruiters to efficiently discover and evaluate suitable talents.

By critically reviewing previous studies and existing platforms, this chapter provides the theoretical and technological foundation for the system design, matching mechanisms, and implementation presented in later chapters.

2.2 Talent Marketplace Systems

Talent marketplace systems are digital platforms that connect individuals with organizations seeking specialized skills and professional services. These platforms enable users to create professional profiles, upload portfolios, search for opportunities, and communicate directly with recruiters.

The rapid growth of digital technologies has significantly transformed recruitment processes. Traditional hiring methods relied heavily on physical applications, paper-based portfolios, and face-to-face interviews. Modern digital platforms have improved recruitment by reducing geographical barriers, increasing accessibility, and providing faster communication between employers and applicants.

Research by Kumar and Lee (2020) suggests that successful digital talent marketplaces generally consist of three core components: user profile management, intelligent search capabilities, and communication mechanisms. Together, these components improve recruitment efficiency while expanding employment opportunities across international markets.

Search and filtering capabilities are particularly important in talent marketplaces because recruiters often need to identify candidates according to specific requirements. Filtering users based on characteristics such as location, experience, skills, and professional category can reduce the number of unsuitable candidates that require manual evaluation.

Although general talent marketplace systems perform effectively for conventional recruitment, they rarely address the specialized requirements of the modeling, entertainment, and pageant industries. Professional modeling recruitment requires visual portfolio management, appearance-based filtering, verification processes, and structured casting workflows, which are generally absent from conventional employment platforms.

Therefore, a specialized talent marketplace can extend the concepts of existing digital recruitment systems by combining professional profiles, multimedia portfolios, advanced search, casting management, and algorithm-based talent matching within a single platform.

2.3 Casting Management Systems in the Modeling Industry

Casting management systems are specialized recruitment platforms designed for selecting models, actors, performers, and contestants for fashion events, advertising campaigns, television productions, films, and beauty pageants.

Traditional casting procedures continue to depend heavily on modeling agencies, manual portfolio reviews, and physical auditions (Nikolaou, 2021). While these approaches allow recruiters to evaluate candidates personally, they require considerable time, financial resources, and administrative effort. Furthermore, aspiring models without agency representation often struggle to gain exposure to recruiters.

Several online platforms, including Model Mayhem and StarNow, provide digital portfolios and casting notices. These systems improve accessibility compared to traditional recruitment; however, they remain focused primarily on portfolio presentation rather than complete casting management.

Critical evaluation of these platforms reveals several limitations:

- Limited support for structured casting workflows.
- Lack of standardized portfolio verification.
- Insufficient search filters for physical characteristics and modeling categories.
- Weak collaboration between recruiters, agencies, photographers, and event organizers.
- Minimal automation during candidate selection.
- Limited algorithm-based matching between casting requirements and candidate profiles.

Consequently, recruiters frequently perform manual candidate comparisons. This can increase recruitment time and make the selection process dependent on individual recruiter effort. A structured matching mechanism can support this process by comparing casting requirements with relevant candidate attributes and producing a ranked list of potentially suitable talents.

2.4 Digital Marketplaces

Digital marketplace platforms such as LinkedIn, Fiverr, and Upwork have transformed professional networking and freelance employment by enabling individuals to market their skills globally.

These platforms generally include user profile management, job posting, messaging systems, search functions, and reputation mechanisms. According to Lukac and Grow (2021), these components contribute significantly to digital workforce expansion by simplifying employer-employee interactions.

Search and recommendation features are also important within digital marketplaces because users may have difficulty manually reviewing a large number of available profiles or opportunities. Structured filtering and recommendation mechanisms can help users identify relevant results more efficiently.

Despite their success, these platforms are designed for general professional services rather than modeling recruitment. They lack industry-specific capabilities such as:

- * Digital casting management.
- * Professional portfolio verification.
- * Appearance-based talent searching.
- * Beauty pageant management.
- * Fashion event coordination.
- * Specialized talent-to-casting matching.

Therefore, while existing digital marketplaces demonstrate the effectiveness of online recruitment, they cannot fully satisfy the operational requirements of modeling and pageant organizations.

2.5 Portfolio-Based Professional Systems

Digital portfolios have become essential tools for professionals working within creative industries. Unlike traditional résumés, portfolios provide visual evidence of an individual's skills, experience, achievements, and professional quality.

Within the modeling industry, portfolios commonly include:

- * Professional photographs.
- * Runway videos.
- * Commercial advertisements.
- * Physical measurements.
- * Awards and achievements.
- * Previous modeling experience.

According to Geyik et al. (2018), digital portfolios can support recruiter decision-making by presenting more comprehensive information about candidates and their capabilities.

However, many existing portfolio platforms suffer from several weaknesses:

- * Lack of standardized portfolio structures.
- * Limited support for multimedia content.
- * Absence of identity verification.
- * Inconsistent presentation formats.
- * Difficulty comparing candidates objectively.

These shortcomings can reduce recruitment efficiency and make it difficult for recruiters to compare multiple candidates consistently.

For the proposed system, portfolio information can also contribute to the talent matching process. Candidate attributes such as experience, skills, location, category, and portfolio information can be considered when determining the suitability of a candidate for a particular casting opportunity.

2.6 Multi-Role User Management

Modern online platforms often serve multiple categories of users, each requiring different permissions and responsibilities. Role-Based Access Control (RBAC) is widely adopted to ensure users can only access functions relevant to their assigned roles (Sandhu et al., 1996).

The proposed system supports multiple stakeholders, including:

- Models.
- Modeling agencies.
- Casting directors.
- Pageant organizers.
- Photographers.
- Fashion designers.
- Administrators.

Each stakeholder performs distinct activities within the platform. For example, models manage their profiles and portfolios, recruiters publish casting opportunities and evaluate applicants, photographers collaborate with talent, while administrators oversee system operations and verification processes.

Implementing RBAC improves security, protects sensitive information, and enhances overall system organization. It also provides a structured foundation for managing different workflows within a single platform.

2.7 Recommendation and Matching Systems

Recommendation systems have become fundamental components of modern digital platforms by assisting users in identifying relevant products, services, or opportunities.

Ricci et al. (2015) categorize recommendation techniques into:

- * Content-based filtering.
- * Collaborative filtering.
- * Hybrid recommendation.

Content-based approaches recommend items according to characteristics associated with the user's requirements or previous preferences. Collaborative approaches use interactions or preferences from multiple users, while hybrid approaches combine different recommendation techniques.

Within talent marketplace platforms, recommendation and matching mechanisms can analyze multiple candidate characteristics, including:

- Physical attributes.
- Professional experience.
- Skills.
- Portfolio quality.
- Geographic location.
- Previous work history.

In a modeling and casting environment, matching can be performed by comparing the requirements of a casting opportunity with the characteristics of available talents. For example, a casting opportunity may specify a particular location, age range, height range, experience level, category, or required skills. A matching algorithm can compare these requirements with candidate profiles and determine which candidates satisfy the specified conditions.

A rule-based or weighted matching approach can provide a suitability score based on the degree to which a candidate satisfies the casting requirements. Candidates with higher compatibility can then be presented earlier in the search or recommendation results. This approach can reduce the amount of manual comparison required from recruiters.

However, the use of matching algorithms must be carefully designed because different casting requirements may have different levels of importance. The algorithm should therefore consider the relevant attributes and apply clearly defined matching criteria rather than making unsupported assumptions about candidate suitability.

Although recommendation algorithms have been widely adopted in e-commerce and entertainment platforms, their application within modeling recruitment remains relatively limited. Most existing casting platforms continue to rely heavily on manual searches performed by recruiters.

Therefore, integrating algorithm-based matching into a casting management platform represents an opportunity to improve candidate discovery, reduce recruitment effort, and provide more structured recommendations to recruiters.

2.8 Advanced Search and Filtering in Talent Recruitment

Search and filtering mechanisms are important components of digital recruitment systems because recruiters may need to evaluate a large number of candidate profiles.

Traditional manual recruitment requires recruiters to individually review candidate information before identifying suitable applicants. Advanced filtering can reduce this workload by allowing recruiters to specify relevant criteria before viewing search results.

For a modeling and talent marketplace, relevant filtering criteria may include:

- Country or location.
- Age.
- Height.
- Professional category.
- Experience.
- Skills.
- Portfolio availability.
- Previous work experience.

A structured filtering mechanism can first eliminate candidates who do not satisfy mandatory requirements. The remaining candidates can then be ranked according to additional matching criteria.

Search and filtering therefore complement algorithm-based matching. Filtering can identify candidates who meet basic requirements, while a matching algorithm can further evaluate their overall suitability and generate a ranked result.

This combination can provide recruiters with a more efficient method of discovering suitable talents while maintaining control over the selection criteria.

2.9 Web-Based System Architecture

Modern web applications typically follow layered client-server architectures that separate the presentation layer, business logic, and database management system.

Pressman and Maxim (2020) argue that layered architectures improve maintainability, scalability, and software quality by separating application responsibilities.

For global talent marketplace systems, scalable architectures are particularly important because they must support:

- Large user populations.
- High-resolution image storage.
- Video portfolio management.
- Real-time messaging.
- Concurrent recruiter activities.
- Search and matching operations.

Cloud computing technologies further improve scalability by providing elastic storage and computing resources while reducing infrastructure costs.

A suitable architecture is therefore important for the proposed system because the platform must manage user accounts, multimedia portfolios, casting opportunities, applications, search operations, matching calculations, and communication between different user roles.

2.10 Technologies Used in Existing Platforms

Recent marketplace applications commonly employ modern web technologies to improve system performance and user experience.

Frontend technologies:

- React.js.
- Angular.
- Vue.js.

Backend technologies:

- Node.js.
- Laravel.
- Django.

Database technologies:

- MySQL.
- PostgreSQL.
- MongoDB.

Cloud services:

- AWS S3.
- Firebase Storage.
- Google Cloud Storage.

Pressman and Maxim (2020) note that modern technologies support scalable full-stack application development and facilitate secure authentication, efficient data management, communication, and multimedia handling.

However, technology alone does not guarantee suitability for specialized industries. Most existing implementations continue to target general-purpose recruitment rather than modeling and casting workflows.

The proposed system therefore combines modern web technologies with specialized functionality such as role-based access control, multimedia portfolio management, advanced search and filtering, casting management, and algorithm-based talent matching.

Comparative Analysis of Existing Systems

Platform	Strengths	Limitations
LinkedIn	Professional networking, job search	Not designed for modeling or casting
Model Mayhem	Portfolio management, casting notices	Limited workflow automation and verification
StarNow	Talent discovery	Weak recommendation and filtering capabilities
Fiverr	Global freelance marketplace	No casting or portfolio verification
Upwork	Secure freelance recruitment	Unsuitable for entertainment industry recruitment

The comparison indicates that although existing platforms successfully support professional networking and freelance recruitment, none provide an integrated environment combining talent management, portfolio verification, intelligent matching, casting workflows, and pageant management.

2.11 Research Gap

The literature review identifies several significant research gaps in existing talent marketplace, recruitment, portfolio, and casting management systems.

First, existing platforms do not comprehensively integrate models, agencies, photographers, pageant organizers, and casting directors within a single collaborative global platform. Although general digital marketplaces provide professional networking and recruitment functions, they are not specifically designed to address the requirements of the modeling, fashion, entertainment, and pageant industries.

Second, existing systems provide limited support for structured casting workflow management. A complete casting process may involve publishing a casting opportunity, defining requirements, discovering suitable talents, reviewing applications, comparing candidates, and selecting suitable applicants. These activities are often distributed across different platforms or handled manually.

Third, algorithm-based talent matching and recommendation remain underutilized within specialized modeling and casting platforms. Existing systems commonly depend on manual searches and portfolio reviews. There is an opportunity to use structured matching algorithms to compare casting requirements with talent attributes such as location, age, height, experience, skills, category, and portfolio information. This can assist recruiters in identifying and ranking potentially suitable candidates more efficiently.

Fourth, advanced search and filtering are not sufficiently integrated with casting workflows. Recruiters may need to manually search through large numbers of candidate profiles even when specific requirements are available. Combining structured filtering with algorithm-based matching could reduce the number of unsuitable candidates presented to recruiters.

Fifth, portfolio verification and standardized talent classification are insufficiently supported in many existing platforms. This can make candidate evaluation inconsistent and increase the effort required from recruiters when comparing different profiles.

Finally, current platforms primarily address general recruitment, freelance employment, or portfolio presentation and do not fully satisfy the specialized operational requirements of modeling, fashion, entertainment, and pageant industries.

These limitations justify the development of the proposed Global Talent Marketplace and Casting Management System. The proposed system aims to address these gaps by integrating multi-role profiles, multimedia portfolios, casting management, advanced search and filtering, algorithm-based talent matching, basic recommendations, and communication within a unified web-based platform.

2.12 Conceptual Framework

The proposed study adopts an Input, Process, Output (IPO) conceptual framework. The framework illustrates how information provided by users and recruiters is processed through the proposed system to produce useful recruitment and casting outcomes.

Input

The major inputs to the proposed system include:

- User profiles.
- Professional portfolios.
- Casting requirements.
- Recruiter preferences.
- Talent attributes.
- Search and filtering criteria.

Process

The system processes the collected information through several interconnected functions:

- User authentication and role management.
- Profile and portfolio management.
- Advanced search and filtering.
- Casting opportunity management.
- Requirement analysis.
- Algorithm-based talent matching.
- Talent ranking and recommendation.
- Application management.
- Communication between users.
- Administrative and verification processes.

During the matching process, the system compares relevant talent attributes with the requirements of a casting opportunity. The matching mechanism can then determine the suitability of available talents and assist in producing ranked results. This process is intended to reduce manual candidate comparison and improve the efficiency of talent discovery.

Output

The expected outputs of the proposed system include:

- Suitable talent recommendations.
- Ranked talent search results.
- Successful talent-to-casting matching.
- More efficient casting decisions.
- Improved professional networking.
- Improved recruitment efficiency.
- Better access to international opportunities.

The conceptual framework demonstrates how user and casting information is transformed through integrated digital processes into meaningful recruitment and casting outcomes.

2.13 Chapter Summary

This chapter critically reviewed existing literature related to talent marketplace systems, casting management platforms, digital marketplaces, portfolio-based applications, recommendation and matching systems, multi-role user management, advanced search and filtering, and modern web technologies.

The review showed that although existing platforms have significantly improved digital recruitment and professional networking, they remain largely focused on general employment, freelance services, or portfolio presentation. Specialized requirements of the modeling, fashion, entertainment, and pageant industries are not comprehensively addressed.

Several gaps were identified, including limited integration of industry stakeholders, incomplete casting workflows, limited algorithm-based talent matching, insufficient advanced search and filtering, inconsistent portfolio verification, and limited integration between candidate profiles and casting requirements.

The proposed Global Talent Marketplace and Casting Management System aims to address these gaps by integrating multi-role user management, professional portfolio management, casting operations, advanced search and filtering, algorithm-based talent matching, basic recommendation, and communication into a unified web-based platform.

The literature review and identified research gaps provide the theoretical and technological foundation for the system methodology, architecture, algorithm design, implementation, and evaluation presented in the following chapter.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter describes the research methodology adopted for the design, development, implementation, and evaluation of the proposed Global Multidimensional Talent Marketplace and Casting Management System. It explains the research approach, software development methodology, system design techniques, algorithm design, development tools, data collection methods, testing procedures, deployment strategy, and security considerations.

A systematic methodology is essential to ensure that the developed system satisfies the identified user requirements while maintaining high standards of quality, scalability, usability, and security. The methodology also provides a structured framework that guides the entire software development process from requirement gathering to system design, implementation, testing, evaluation, deployment, and maintenance.

3.2 Research Approach

This study adopts the Design and Development Research (DDR) methodology. Design and Development Research focuses on designing, developing, and evaluating technological solutions that address practical problems while contributing to knowledge within a specific domain.

The selected research approach consists of the following stages:

- Problem identification and analysis.
- Requirement gathering.
- System design.
- Algorithm design.
- Prototype development.
- System implementation.
- Testing and evaluation.
- Deployment and maintenance.

The Design and Development Research methodology is appropriate because the objective of this study is to develop a functional web-based application that addresses the inefficiencies of existing casting and talent recruitment systems. The methodology also supports the development and evaluation of the proposed search, filtering, and algorithm-based talent matching mechanisms.

3.3 Software Development Methodology

The software was developed using the Agile Software Development Methodology, following an iterative and incremental development process. Agile was selected because it supports continuous improvement, frequent testing, user feedback, and flexible adaptation to changing requirements.

Each development iteration consisted of planning, implementation, testing, review, and refinement. This enabled features to be developed independently while ensuring that feedback obtained during development could be incorporated into subsequent iterations.

The Agile lifecycle adopted in this study consists of six major phases:

- Requirement Analysis.
- System Design.
- System Implementation.
- Testing.
- Deployment.
- Maintenance.

The Agile approach was particularly suitable for this project because the proposed system contains several independent but interconnected modules, including user management, portfolio management, casting management, search and filtering, algorithm-based talent matching, messaging, and administration. These modules could be developed and tested incrementally rather than requiring the complete system to be implemented at once.

3.3.1 Requirement Analysis

The first phase involved identifying functional and non-functional requirements through literature review, observation of industry practices, competitor analysis, and informal discussions with potential users.

The analysis focused on understanding the operational workflow of the modeling and entertainment industry, including casting directors, modeling agencies, photographers, pageant organizers, and professional models. Particular attention was given to the processes involved in talent discovery, portfolio evaluation, casting management, candidate selection, and communication between industry stakeholders.

The functional requirements identified include:

- User registration and authentication.
- Role-based user management.
- Talent profile management.
- Portfolio creation and media uploads.
- Casting call creation and management.
- Application management.
- Advanced talent search and filtering.
- Algorithm-based talent-to-casting matching.
- Talent recommendation and ranking.
- Messaging between users.
- Administrative management and verification.

The non-functional requirements include:

- Security.
- Performance.
- Scalability.
- Reliability.
- Availability.
- Usability.
- Maintainability.

The identified requirements served as the foundation for the subsequent system design, algorithm design, implementation, and testing phases.

3.3.2 System Design

Following requirement analysis, the overall architecture, database structure, system functionality, and user interactions of the proposed system were designed using Object-Oriented Design (OOD) principles.

Several software engineering models were developed to represent the system structure and behavior. The design artifacts include:

- Use Case Diagram.
- Class Diagram.
- Entity Relationship Diagram (ERD).
- Sequence Diagrams.
- Activity Diagrams.
- Database Schema.
- System Architecture Diagram.

The system design also considers the interaction between the search, filtering, casting, and algorithm-based matching components. These components are designed to work together so that recruiters can define casting requirements, identify suitable talents, and review ranked results.

The application adopts a Three-Tier Architecture, consisting of the following layers:

Presentation Layer

The presentation layer provides the graphical user interface through which users interact with the application. It allows different user roles to manage profiles, portfolios, casting opportunities, applications, searches, recommendations, and communication.

Application Layer

The application layer contains the business logic responsible for authentication, role-based access control, profile and portfolio management, casting management, search and filtering, algorithm-based talent matching, recommendation services, messaging, and other system operations.

The matching component processes relevant talent attributes and casting requirements to determine candidate suitability according to the defined matching criteria.

Data Layer

The data layer manages persistent storage of user profiles, portfolios, casting information, applications, matching-related information, and administrative records.

This architectural separation improves maintainability, scalability, security, and software quality while allowing individual system components to be developed, tested, and maintained independently

3.4 System Analysis and Requirements

System analysis was conducted to identify the functional and non-functional requirements of the proposed platform. The requirements were derived from the identified problems, literature review, and expected needs of the different stakeholder groups.

The main stakeholders considered in the system include:

- Models and talents.
- Modeling agencies.
- Casting directors.
- Pageant organizers.
- Photographers.
- Fashion designers.
- System administrators.

The requirements analysis focused on the major functions required to support profile creation, portfolio management, casting opportunities, talent discovery, candidate matching, applications, communication, and administrative management.

3.4.1 Functional Requirements

The proposed system includes the following major functional requirements:

- User registration and authentication.
- Role-based user management.
- Profile creation and management.
- Professional portfolio management.
- Image and video portfolio uploads.
- Casting opportunity creation and management.
- Talent search and filtering.
- Algorithm-based talent matching.
- Talent recommendation and ranking.
- Casting application management.
- Communication between relevant users.
- Administrative management and verification.

3.4.2 Non-Functional Requirements

The major non-functional requirements include:

- Security: User information and system resources should be protected through appropriate authentication, authorization, validation, and secure data handling.
- Performance: Search, filtering, profile loading, and matching operations should provide acceptable response times.
- Scalability: The architecture should support increasing numbers of users, profiles, portfolios, and casting opportunities.
- Usability: The interface should be understandable and easy to navigate for different categories of users.
- Reliability: The system should perform consistently and handle errors appropriately.
- Maintainability: The system should use a structured architecture that allows future modifications and feature improvements.
- Availability: The deployed platform should remain accessible to authorized users when required.

3.5 System Design

The system design phase defines the structure, behaviour, data flow, and interaction between the components of the proposed platform.

Object-Oriented Design (OOD) principles are used to represent the major entities and functions of the system. The system design includes:

- Use Case Diagram.
- Class Diagram.
- Entity Relationship Diagram (ERD).
- Sequence Diagrams.
- Activity Diagrams.
- Database Design.
- System Architecture.
- User Interface Design.

The design process considers the different user roles and their permissions while ensuring that the modules communicate effectively with each other.

3.6 Algorithm Design

Algorithm design is an important component of the proposed system because the platform aims to improve talent discovery and casting efficiency through structured search, filtering, matching, and recommendation mechanisms.

The algorithms are designed to compare information stored in talent profiles with requirements specified by recruiters or casting organizers. The purpose is to reduce unnecessary manual candidate comparison and provide recruiters with more relevant results.

The proposed algorithms include:

3.6.1 Talent Search and Filtering Algorithm

The search and filtering algorithm allows recruiters to specify criteria such as:

- Country or location.
- Age.
- Height.
- Professional category.
- Experience.
- Skills.
- Portfolio availability.

The system processes the selected criteria and retrieves talent profiles that satisfy the specified conditions.

The general process is:

1. Receive the recruiter's search criteria.
2. Retrieve relevant talent profiles from the database.
3. Compare each profile with the selected criteria.
4. Exclude profiles that do not satisfy mandatory conditions.
5. Return the profiles that satisfy the selected filters.

6. Display the results to the recruiter.

3.6.2 Talent-to-Casting Matching Algorithm

The talent-to-casting matching algorithm compares the requirements of a casting opportunity with the attributes of available talents.

Potential matching attributes include:

- Location.
- Age.
- Height.
- Experience.
- Professional category.
- Skills.
- Portfolio information.

The general process is:

1. Retrieve the requirements of a casting opportunity.
2. Retrieve eligible talent profiles.
3. Compare the required attributes with each talent's attributes.
4. Determine the degree of compatibility.
5. Calculate a suitability score based on the defined matching criteria.
6. Rank candidates according to their matching results.
7. Present the most relevant candidates to the recruiter.

The algorithm is intended to support recruiter decision-making rather than completely replace human selection. Final candidate selection remains the responsibility of the relevant recruiter or casting organizer.

3.6.3 Talent Recommendation Algorithm

The recommendation mechanism uses available profile and casting information to identify potentially relevant opportunities or candidates.

For example, when a recruiter creates a casting opportunity, the system can identify talents whose profile attributes are compatible with the requirements of that opportunity. Similarly, talent users can be presented with casting opportunities that match their profile information.

The recommendation process consists of:

1. Collect relevant user or casting information.
2. Identify matching attributes.
3. Compare the available profiles or opportunities.
4. Determine compatibility.
5. Rank the relevant results.
6. Present recommendations to the user.

The recommendation mechanism is designed as a structured algorithm-based approach rather than claiming advanced artificial intelligence or machine-learning functionality.

3.7 System Architecture

The proposed system follows a three-tier web application architecture consisting of:

- Presentation Layer.
- Application/Business Logic Layer.
- Data Layer.

The presentation layer provides the user interface through which users interact with the platform. The application layer handles authentication, business rules, casting management, search, filtering, matching, recommendations, and communication. The data layer manages user profiles, portfolios, casting information, applications, and other system data.

This separation improves maintainability, scalability, and organization of the application.

3.8 System Implementation

The proposed system is implemented using modern web technologies.

Frontend

- React.js.
- HTML5.
- CSS3.
- JavaScript.
- Bootstrap or Tailwind CSS.

Backend

- Node.js.
- Express.js.

Database

- MongoDB.

Main System Modules

The implementation consists of the following major modules:

1. Authentication and Authorization.
2. User Management.
3. Profile Management.
4. Portfolio Management.
5. Casting Management.
6. Search and Filtering.
7. Talent Matching and Recommendation.
8. Messaging.
9. Administration.

Each module is developed and tested incrementally following the Agile development approach.

3.9 Data Collection Methods

Data required for system development and evaluation is collected through appropriate research and development activities.

The main sources include:

- Literature and previous research studies.
- Analysis of existing talent marketplace and casting platforms.
- User requirements.
- Stakeholder feedback.

- System testing results.
- User evaluation and feedback.

The collected information is used to refine system requirements, interface design, functionality, and evaluation criteria.

3.10 Testing and Evaluation

Testing is conducted to determine whether the developed system satisfies its functional and non-functional requirements.

The testing strategy includes:

Unit Testing

Individual functions and components are tested independently to verify that they operate correctly.

Integration Testing

Interactions between different modules are tested to ensure that data and functionality are correctly integrated.

System Testing

The complete system is tested to determine whether all major functions operate together as expected.

User Acceptance Testing

Selected users evaluate the system to determine whether it satisfies practical requirements and provides an acceptable user experience.

Matching and Recommendation Evaluation

The search, matching, and recommendation mechanisms are evaluated to determine whether they return relevant results based on the defined criteria.

The evaluation should use actual test cases and collected results rather than assumed or fabricated results.

3.11 Deployment Strategy

The proposed system uses cloud-based deployment technologies.

The frontend can be deployed using Vercel, while the backend can be deployed using Render. MongoDB Atlas can be used to provide cloud-based database services.

This deployment approach provides accessibility through the internet and supports the scalability requirements of a global web-based platform.

3.12 Security Considerations

Security is an important consideration because the proposed system manages user accounts, personal profiles, portfolios, applications, and communication.

The system incorporates security mechanisms including:

- Password hashing using bcrypt.
- JSON Web Token (JWT) based authentication.
- Role-Based Access Control (RBAC).

- Input validation.
- Secure file-upload handling.
- HTTPS communication.
- Session and authentication management.
- Protection against common injection attacks.
- Protection against cross-site scripting (XSS).

These mechanisms are intended to protect user information and prevent unauthorized access to system functions.

3.13 Chapter Summary

This chapter presented the research and software development methodology adopted for the proposed Global Multidimensional Talent Marketplace and Casting Management System.

The study follows Design and Development Research and uses Agile software development to support iterative implementation and continuous refinement. The chapter also described system requirements, system design, algorithm design, architecture, implementation technologies, data collection, testing, deployment, and security considerations.

Particular attention was given to the proposed search, filtering, talent-to-casting matching, and recommendation algorithms, which are intended to improve talent discovery and casting efficiency. The next chapter presents the implementation and evaluation of the developed system.

CHAPTER 4: RESULTS

4.1 Introduction

This chapter presents the results obtained from the design, development, implementation, and testing of the proposed Global Multidimensional Talent Marketplace and Casting Management System. The purpose of the developed system is to provide a centralized digital platform that connects models, aspiring talents, modeling agencies, photographers, casting directors, pageant organizers, and other industry professionals.

The results presented in this chapter are based on the functional features implemented in the prototype, the system architecture described in Chapter 3, and the testing activities conducted during development. The evaluation focuses on the main functional requirements, usability, system performance, security, role-based access, portfolio management, casting management, search and filtering, messaging, and administrative functionality.

The developed system follows a three-tier architecture consisting of the presentation layer, application layer, and data layer. The implementation uses React.js, HTML5, CSS3, and JavaScript for the frontend, Node.js and Express.js for the backend, and MongoDB for database management.

4.2 System Implementation Results

The development process resulted in a functional web-based prototype capable of supporting the main activities required in a digital talent marketplace. The implementation transformed the approved system design into an interactive application that supports different user roles and recruitment-related activities.

The major modules implemented in the system are:

- Authentication Module
- User Management Module
- Portfolio Management Module

- Casting Management Module
- Search and Filtering Module
- Messaging Module
- Administration Module

These modules were developed according to the functional requirements identified during the requirement analysis phase.

4.2.1 Authentication and User Management

The authentication module allows users to register and access the system securely. The system supports different categories of users, allowing functionality to be provided according to the responsibilities of each role.

Role-based access control was incorporated to restrict users from accessing functions that are not relevant to their assigned roles. This supports the multi-role nature of the proposed marketplace and improves system security and organization.

The implemented security mechanisms include password hashing using bcrypt, JSON Web Token authentication, role-based access control, input validation, secure file upload validation, HTTPS communication, and session management.

4.2.2 Talent Profile and Portfolio Management

The system provides users with the ability to create and manage professional profiles. Talent users can provide relevant professional information and develop portfolios containing multimedia content.

Portfolio management is an important component because modeling and entertainment recruitment depends heavily on visual evidence such as photographs, videos, previous work, measurements, and achievements.

The implemented portfolio functionality provides a more structured method of presenting professional information compared with relying solely on social media profiles.

4.2.3 Casting Management

The casting management module enables industry professionals and organizers to create casting opportunities. Talent users can view available opportunities and submit applications.

The system supports the basic casting workflow from the publication of a casting opportunity to application submission and candidate management. This addresses one of the major problems identified in the research, where existing recruitment methods often depend on manual communication, social media posts, emails, and informal networks.

4.2.4 Search and Filtering

The search and filtering module provides recruiters with a mechanism to identify suitable talents based on relevant profile information.

The proposed system considers attributes such as age, height, experience, location, category, and portfolio information. This functionality is intended to reduce the amount of time recruiters spend manually reviewing large numbers of profiles.

The implementation therefore supports the research objective of introducing advanced search and filtering mechanisms into a specialized talent recruitment environment.

4.2.5 Messaging

The messaging module provides communication functionality between users. It allows talent and industry professionals to communicate through the platform rather than depending entirely on external communication channels.

This feature contributes to the objective of creating a centralized environment in which recruitment-related interactions can occur.

4.2.6 Administration

The administration module provides system-level management functionality. Administrators can oversee system operations and support moderation and management activities.

The inclusion of an administrative role is important because a global talent marketplace requires mechanisms for managing users, maintaining platform quality, and supporting secure system operation.

4.3 Functional Testing Results

Functional testing was conducted to determine whether the implemented features operated according to their expected requirements. The testing process included unit testing, integration testing, system testing, and User Acceptance Testing.

Unit testing was used to test individual modules independently. Integration testing was used to verify communication between the frontend, backend, and database. System testing evaluated the complete application, while User Acceptance Testing was intended to determine whether the system satisfied practical requirements of potential users.

4.4 Integration Testing Results

Integration testing was performed to verify communication between the frontend, backend, and database components.

The frontend provides the user interface, while the backend handles business logic, authentication, casting management, messaging, and system operations. The database stores information related to users, portfolios, casting calls, applications, and administrative records.

The successful completion of the integration testing indicates that the major application components were able to communicate as required during the documented test scenarios.

This is particularly important for the proposed system because recruitment activities require several components to work together. For example, a casting application requires the user interface to communicate with the backend and subsequently store the application information in the database.

4.5 Security Testing Results

Security was considered throughout the development process. The testing process included role-based access testing and validation of security-related functionality.

The role-based access test successfully restricted access according to user permissions. This demonstrates that the implemented access control mechanism supports the requirement for multiple user roles.

The system also incorporates bcrypt password hashing, JWT authentication, input validation, secure file upload validation, HTTPS communication, and protection against common web vulnerabilities such as injection and Cross-Site Scripting.

These mechanisms provide a foundation for protecting user information and system resources.

4.6 Performance Results

System performance was evaluated as part of the documented testing process. The performance test case assessed whether the system responded within an acceptable time.

The recorded result for the system performance test was “Response time acceptable”, and the test was marked as Pass.

Although the prototype demonstrated acceptable response behaviour during testing, comprehensive large-scale performance benchmarking was outside the scope of the current academic prototype. Therefore, additional testing with a significantly larger number of concurrent users would be appropriate before commercial deployment.

4.7 Deployment Results

Following successful testing, the prototype was deployed using cloud-based services. The deployment configuration specified:

- * Frontend: Vercel
- * Backend: Render
- * Database: MongoDB Atlas

This deployment architecture allows users to access the system through standard web browsers without installing additional software.

The cloud-based approach also provides a foundation for future scalability because the system can potentially accommodate additional users and resources as the platform develops.

4.8 Achievement of Research Objectives

The results demonstrate progress toward the research objectives identified in Chapter 1.

Objective 1: Analyze existing systems and industry practices

Existing platforms and recruitment practices were reviewed and compared. The analysis identified limitations relating to casting workflows, portfolio verification, specialized search, and industry-specific recruitment.

Objective 2: Identify system requirements

Functional and non-functional requirements were identified through literature review, observation, competitor analysis, and informal discussions with potential users.

Objective 3: Design the system architecture

A three-tier architecture was designed consisting of presentation, application, and data layers. Supporting design artifacts included use case diagrams, class diagrams, ER diagrams, sequence diagrams, activity diagrams, database schema, and system architecture diagrams.

Objective 4: Develop a web-based platform

A functional web application was developed using React.js, Node.js, Express.js, and MongoDB.

Objective 5: Implement advanced search and filtering

The system includes a search and filtering module intended to help industry professionals identify suitable talent using profile attributes. The corresponding test case successfully produced relevant search results.

Objective 6: Incorporate casting management

The system provides casting call creation and application management functionality. Both casting creation and application submission were successfully tested.

Objective 7: Ensure usability, accessibility, and scalability

The system was designed as a responsive web-based application and incorporates a scalable architectural structure. However, extensive large-scale scalability testing remains an area for future work.

Objective 8: Evaluate system effectiveness

The prototype was evaluated through functional, integration, system, security, performance, and user acceptance testing activities. The documented functional test cases achieved passing results.

4.9 Chapter Summary

This chapter presented the results of the developed Global Multidimensional Talent Marketplace and Casting Management System. The implemented prototype provides authentication, user management, portfolio management, casting management, search and filtering, messaging, and administrative functionality.

Testing covered the major system functions and non-functional requirements. The documented ten test cases achieved a Pass status, including registration, login, casting creation, casting application, search and filtering, messaging, role-based access, file upload, system performance, and logout.

The results demonstrate that the developed prototype successfully implements the major functional requirements identified during the requirement analysis phase. The following chapter discusses these findings in relation to the research problem, objectives, literature review, and identified research gap.

Create Casting Call - Global Talent

localhost:5173/castings/create

Global Talent

Dashboard

Castings

Applications

Talents

Agencies

Messages

Reports

Settings

Sarah Johnson
Casting Director

Create Casting Call

Basic InfoRequirementsMediaReview

Casting Title *

Summer Fashion Campaign

Type *

Photoshoot

Location *

Los Angeles, USA

Date *

2025-06-15

Description *

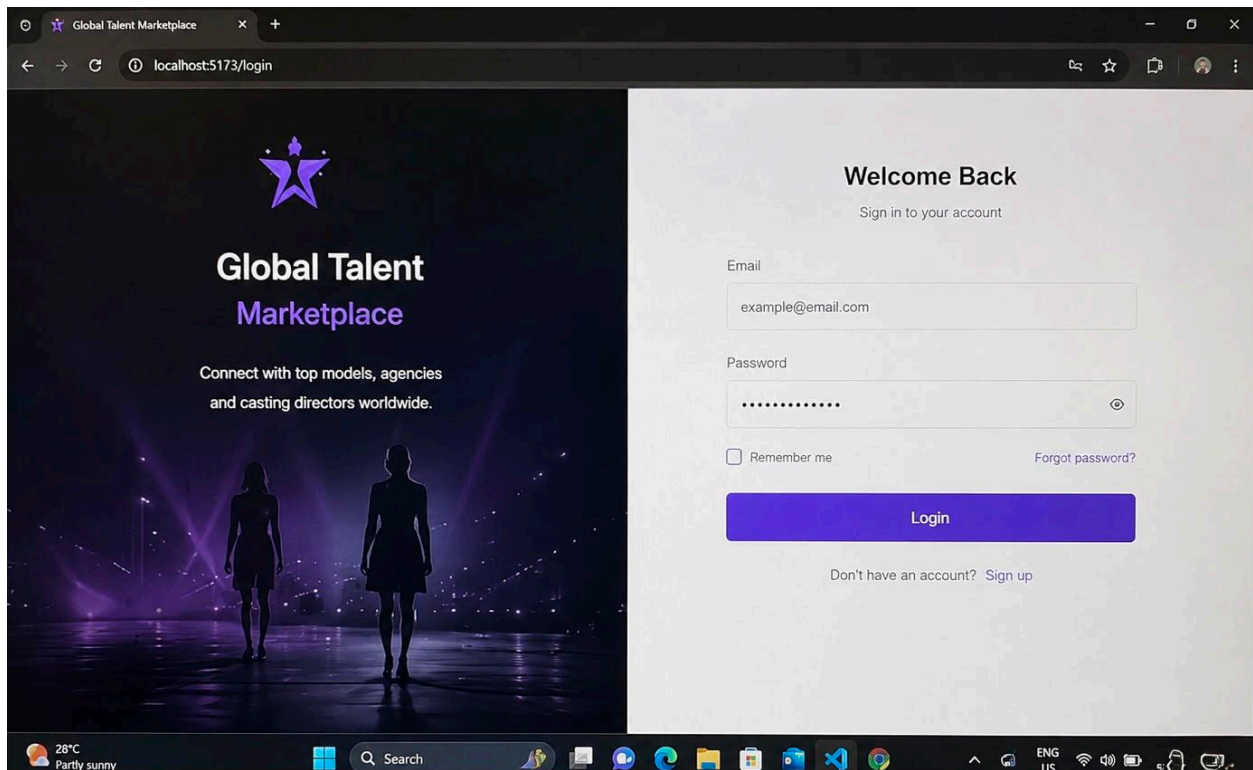
Looking for models for a summer fashion photoshoot for our new collection.

Application Deadline *

2025-06-01

Cancel

Next



CHAPTER 5: DISCUSSION

5.1 Introduction

This chapter discusses the findings presented in Chapter 4 in relation to the research problem, research questions, objectives, literature review, and conceptual framework. The purpose of the discussion is to determine how effectively the developed Global Multidimensional Talent Marketplace and Casting Management System addresses the limitations identified in existing recruitment and casting practices.

The proposed system was developed to provide a centralized digital environment for models, aspiring talents, modeling agencies, photographers, casting directors, pageant organizers, and other industry professionals. The results indicate that the prototype successfully implements the main functions required for digital talent recruitment, including authentication, professional profiles, portfolio management, casting management, search and filtering, messaging, and administration.

5.2 Discussion of the Main Findings

The main finding of the study is that a specialized digital platform can bring several recruitment activities into a single environment. Existing recruitment practices identified during the study commonly depend on social media, personal networks, agencies, email, and manual portfolio reviews. These approaches can make the recruitment process fragmented and time-consuming.

The developed system addresses this issue by combining multiple recruitment-related functions within one platform. Instead of requiring users to depend on separate platforms for professional profiles, portfolios, casting opportunities, applications, search, and communication, these activities can be managed through the proposed system.

This directly supports the overall aim of the research, which was to design and develop a global talent marketplace and casting management system capable of improving recruitment efficiency, transparency, and accessibility.

5.3 Discussion of Authentication and Role-Based Access

One of the important findings is the successful implementation of multi-role authentication and access control.

The proposed platform supports different stakeholders, including models, agencies, casting directors, photographers, pageant organizers, and administrators. Each stakeholder has different responsibilities within the system.

The implementation of Role-Based Access Control is therefore appropriate because users should not have unrestricted access to every system function.

The successful result of the role-based access test demonstrates that the prototype can restrict access according to user roles. This contributes to system security and improves organizational structure.

This finding is consistent with the literature reviewed in Chapter 2, where Role-Based Access Control was identified as an important mechanism for managing permissions within systems involving multiple categories of users.

5.4 Discussion of Talent Profiles and Portfolios

Portfolio management is particularly important within the modeling and creative industries because professional suitability cannot always be represented through traditional text-based resumes.

The literature review identified digital portfolios as an important source of visual evidence relating to professional experience, skills, achievements, and quality. However, existing portfolio systems can have problems relating to standardization, multimedia support, identity verification, and candidate comparison.

The proposed system responds to this requirement by providing structured talent profiles and portfolio management functionality.

This creates a more industry-specific recruitment environment compared with general professional networking platforms. Recruiters can access professional information and portfolio material within the same system, supporting more efficient candidate evaluation.

However, the current system does not implement advanced identity verification or automated talent scoring. These features were explicitly identified as outside the scope of the current research.

5.5 Discussion of Casting Management

Casting management is one of the key areas where the proposed system differs from general recruitment platforms.

Existing digital marketplaces such as LinkedIn, Fiverr, and Upwork provide general recruitment or freelance functionality, but they do not specifically address the complete requirements of modeling and pageant casting. The literature review identified the absence of dedicated casting workflows, appearance-based searching, pageant management, and specialized portfolio verification as limitations.

The developed system introduces casting call creation and application management as dedicated features.

The successful results for the casting creation and application test cases demonstrate that the prototype can support the basic digital casting workflow.

This finding supports the research gap identified in Chapter 2, which highlighted the lack of integrated casting workflow management from opportunity publication to candidate application and selection.

5.6 Discussion of Search and Talent Discovery

Efficient talent discovery is another major requirement of the proposed system.

Recruiters may need to evaluate large numbers of potential candidates based on characteristics such as location, experience, category, age, height, and portfolio information. Manual searching through social media and separate portfolio platforms can increase recruitment time.

The proposed search and filtering functionality addresses this issue by providing a structured method of locating relevant talent.

The successful Search and Filter test indicates that the system was able to display relevant results during testing. This supports the objective of reducing the effort required to identify suitable candidates.

The literature review also identified recommendation and matching systems as an opportunity for improving recruitment efficiency. Content-based, collaborative, and hybrid recommendation techniques have been widely studied in other digital platforms, while their application in modeling recruitment remains more limited.

The current prototype provides basic matching/search functionality rather than a fully developed artificial-intelligence recommendation engine. Therefore, intelligent recommendation remains an opportunity for future development.

5.7 Discussion of Communication

Communication is an important component of recruitment because recruiters and talent need to exchange information regarding casting opportunities, applications, and professional activities.

The implemented messaging module provides an internal communication mechanism. The successful messaging test demonstrated that messages could be sent and received.

This contributes to the goal of reducing dependence on external communication channels and creating a centralized recruitment environment.

However, additional features such as advanced notifications, real-time communication, file sharing, and video communication could further improve the system in future versions.

5.8 Discussion of Security

Security is particularly important because the proposed system stores user profiles, professional portfolios, applications, and other information.

The implemented security mechanisms include password hashing, JWT authentication, role-based access control, input validation, secure file upload validation, HTTPS communication, session management, and protection against common web vulnerabilities.

The successful role-based access test provides evidence that access restrictions are functioning within the tested prototype.

Nevertheless, security testing in a prototype environment cannot guarantee complete protection against every possible cybersecurity threat. A commercial version would require more comprehensive penetration testing, vulnerability assessment, monitoring, logging, backup strategies, and security auditing.

5.9 Discussion of System Performance and Scalability

The documented performance test produced an acceptable response-time result and was marked as Pass.

The system's three-tier architecture also provides a suitable foundation for future scalability because the presentation, business logic, and data management responsibilities are separated. This separation can make the system easier to maintain and extend.

Cloud deployment using Vercel, Render, and MongoDB Atlas further provides a foundation for global accessibility and future expansion.

However, the current research was conducted within a prototype and academic environment. The study acknowledges limitations such as a limited development period, limited financial resources, limited real-world industry datasets, and evaluation using a relatively small sample of users.

Therefore, the current results should be interpreted as evidence of prototype functionality rather than proof of performance at global commercial scale.

5.10 Relationship Between Findings and Research Questions

Research Question 1

What technical and professional challenges do models, recruiters, and pageant organizers face when using existing recruitment methods?

The study identified several challenges, including fragmented recruitment, dependence on social media, difficulty verifying professional experience, limited international opportunities, manual portfolio reviews, inadequate pageant recruitment functions, and limited communication tools.

The developed system addresses several of these challenges through centralized profiles, portfolios, casting calls, search and filtering, applications, messaging, and administrative management.

Research Question 2

How can a secure multi-role software architecture improve interactions among different stakeholder groups?

The implementation demonstrates that role-based access can separate system functionality according to stakeholder responsibilities. Models, recruiters, photographers, organizers, and administrators can operate within the same platform while having different permissions.

This creates a more organized and secure environment for interaction.

Research Question 3

How can country-based talent discovery improve international recruitment?

The proposed system includes country and location-related talent discovery as part of the search and filtering concept. This can reduce geographical barriers by allowing recruiters to identify talent from different locations.

However, the current prototype does not provide sufficient empirical evidence to measure the real-world international recruitment impact quantitatively.

Research Question 4

How can profile verification reduce fraudulent recruitment activities?

The research identifies profile verification as an important requirement because fake profiles and fraudulent casting opportunities can reduce trust. The proposed system provides a structured professional profile environment and security mechanisms, but full legal identity verification and background checking are outside the scope of the current research.

Therefore, the system provides a foundation for improved trust but does not completely solve identity fraud.

Research Question 5

How effective is the proposed system in terms of usability, performance, and user satisfaction?

The documented testing indicates that the major functional test cases passed and that the system demonstrated acceptable response performance. However, the current document does not provide quantitative user satisfaction results such as questionnaire scores or statistical analysis.

Therefore, the prototype demonstrates functional effectiveness, while a more comprehensive evaluation of user satisfaction would require additional empirical data.

5.11 Relationship with the Research Gap

The literature review identified several gaps in existing systems, including the absence of a comprehensive platform integrating models, agencies, photographers, pageant organizers, and casting directors; limited structured casting workflows; underuse of intelligent matching; insufficient portfolio verification; and limited support for specialized modeling and pageant recruitment.

The developed prototype addresses several of these gaps by integrating:

- Multiple user roles
- Professional talent profiles
- Portfolio management
- Casting call management
- Casting applications
- Search and filtering
- Messaging
- Administrative management
- Role-based access control

Therefore, the results provide evidence that the proposed system is capable of addressing several limitations identified in existing platforms.

However, some research gaps remain partially addressed. In particular, advanced artificial intelligence matching, facial recognition, automatic talent scoring, legal identity verification, payment functionality, and mobile applications were not included in the current scope.

5.12 Contribution of the Developed System

The developed system contributes to the research area by demonstrating how a specialized digital marketplace can combine recruitment and casting activities within a single web-based environment.

From a technological perspective, the project demonstrates the application of modern full-stack development technologies, including React.js, Node.js, Express.js, MongoDB, cloud hosting, and role-based security mechanisms.

From an industry perspective, the system provides a structured alternative to fragmented recruitment through social media, email, agencies, and manual portfolio review.

From an academic perspective, the study provides a foundation for future research into intelligent talent recommendation, automated matching, digital portfolio verification, and specialized recruitment platforms.

5.13 Limitations

Although the results demonstrate successful prototype functionality, several limitations should be considered.

First, the system was developed as an academic prototype rather than a fully commercial platform. Second, the project was conducted within a limited development period and with limited financial resources. Third, access to real-world industry datasets was limited. Fourth, the evaluation was conducted using a relatively small sample of users.

In addition, advanced AI features, online payments, digital contract signing, legal identity verification, background checks, and mobile application development were excluded from the current scope.

These limitations mean that the findings should primarily be considered evidence of the feasibility and functionality of the proposed system rather than evidence of complete commercial readiness.

5.14 Recommendations for Future Development

Several improvements can be introduced in future versions of the system.

5.14.1 Artificial Intelligence-Based Matching

Future versions could implement intelligent recommendation algorithms to automatically match talent profiles with casting requirements.

5.14.2 Advanced Profile Verification

Identity verification mechanisms could be introduced to improve trust and reduce fraudulent profiles.

5.14.3 Automated Talent Ranking

A ranking mechanism could help recruiters compare candidates based on predefined casting requirements and professional attributes.

5.14.4 Mobile Application

A dedicated mobile application could improve accessibility for models, recruiters, and organizers who frequently use mobile devices.

5.14.5 Online Payments and Contracts

Future versions could incorporate secure payment processing and digital contract management for commercial recruitment activities.

5.14.6 Advanced Analytics

Recruiters and administrators could benefit from dashboards containing recruitment statistics, application trends, talent engagement, and casting performance.

5.14.7 Large-Scale Performance Testing

Future research should evaluate the system using larger datasets and higher numbers of concurrent users to determine its scalability and performance under real-world conditions.

5.15 Chapter Summary

This chapter discussed the results of the Global Multidimensional Talent Marketplace and Casting Management System in relation to the research questions, objectives, literature review, and research gap.

The findings indicate that the developed prototype successfully implements the major functions required for a centralized digital talent marketplace, including authentication, role-based access, professional profiles, portfolio management, casting management, search and filtering, messaging, and administration.

The documented testing results demonstrate successful operation of the major tested functions, with all ten documented test cases achieving a Pass status. The system also demonstrated acceptable response performance during the documented performance test.

The discussion shows that the proposed system addresses several limitations identified in existing recruitment and casting platforms. However, advanced features such as AI-based matching, identity verification, automatic talent scoring, mobile applications, payment processing, and digital contracts remain opportunities for future development.

Overall, the study demonstrates the feasibility of developing a specialized web-based talent marketplace that integrates recruitment and casting activities within a centralized digital environment.

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